reduces the entire study of wave functions and wave equations to a very small number of algebraic tricks, which almost always involve the commutation relations. In fact, whenever you see a pair of operators, my advice is to figure out their commutator. If the commutator is a new operator that you haven't seen before, find its commutator with the original pair. That's when the fun happens.

Obviously, this advice can lead to an unending chain of boring computations. But once in a while you may get lucky and find a set of operators that *close under commutation*. Whenever that happens, you're in business; as we will see, operator methods have tremendous power.

Now, let's apply this approach to our harmonic oscillator. We begin with the Hamiltonian expressed in terms of the operators **P** and **X**:

$$\mathbf{H} = \frac{\mathbf{P}^2 + \omega^2 \mathbf{X}^2}{2}. \qquad (10.17)$$

To figure out the rest of the energy levels, we'll use some tricks. The idea is to cleverly use the properties of **X** and **P** (in particular, the commutation relation [**X**, **P**] = *iℏ*) to construct two new operators, called *creation* and *annihilation* operators. When a creation operator acts on an energy eigenvector (or eigenfunction), it produces a new eigenvector that has the next higher energy level. An annihilation operator does just the opposite: it produces an eigenvector whose energy is one level lower than the energy of the eigenvector it started with. So, roughly speaking, the thing that they create and annihilate is energy. They're also called *raising* and *lowering* operators. But remember: operators act on state vectors, not on systems. To see how these operators work, let's rewrite the Hamiltonian in the form

$$\mathbf{H} = \frac{1}{2}(\mathbf{P}^2 + \omega^2 \mathbf{X}^2). \qquad (10.18)$$

This is a classical as well as a quantum mechanical Hamiltonian, and it would be just as correct to use the lowercase symbols *p* and *x*. However,